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Multi-modal interaction enhanced segment anything model (MIE-SAM) for RGB-T salient object detection

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ABSTRACT

RGB-T Salient Object Detection (RGB-T SOD) effectively leverages the complementary information of RGB images and thermal infrared images to locate important targets in complex environments, such as low light, rainy and foggy weather, or cluttered backgrounds. However, the existing deep-learning based models have two key problems: the fixed fusion strategy can not adapt to the changing environmental conditions, and the scarcity of pixel-level annotation leads to over-fitting. While the Segment Anything Model (SAM) exhibits excellent generalization, its adaptation to RGB-T SOD is challenged by the lack of saliency semantics, RGB-only pre-training and manual prompt dependency. To address these challenges, we propose a Multi-modal Interaction Enhanced Segment Anything Model (MIE-SAM). The framework reconfigures SAM's image encoder into a weight-sharing dual-branch image encoders. A Multi-modal Low-Rank Adaptation module (Mm-LoRA) is embedded in the frozen dual-branch image encoders to inject saliency-specific semantics and facilitate deep cross-modal feature interaction while preserving SAM's pre-trained knowledge via parameter-efficient fine-tuning. Furthermore, a Dynamic Fusion Module (DFM) learns dynamic fusion weights to adaptively aggregate multi-modal embeddings based on their environmental reliability, ensuring robust integration in varying environments. Finally, a Progressive Decoder Module (PDM) directly translates the fused features into the fine-grained saliency map in an entirely prompt-free, end-to-end manner. Extensive experiments on several publicly available RGB-T SOD datasets show that our method achieves state-of-the-art performance, exhibiting robustness and generalization ability in a variety of challenging scenarios. The code is available at <https://github.com/Aazzz66/MIESAM>.

1. Introduction

Salient Object Detection (SOD) aims to automatically identify and segment visually prominent objects from an image. Automatically selecting the most important information for complex visual problems can improve information processing efficiency of downstream vision tasks, such as semantic segmentation (Garcia-Garcia et al., 2018; Hoyer et al., 2019), object tracking (Hong et al., 2015; Li et al., 2019; Lu et al., 2021), medical image segmentation (Wu et al., 2021) and image retrieval (Gao et al., 2015; Shao & Brady, 2006; Zhou et al., 2019). Despite the breakthroughs in RGB SOD (Liu et al., 2019; Zhao et al., 2019), their performance often degrades when dealing with challenging scenes such as low-light conditions, or backgrounds with high textural similarity.

To address these limitations, thermal infrared imaging has been used to provide temperature distribution information. RGB-T SOD (Cong et al., 2022; Tu et al., 2022b; Zhang et al., 2020b) leverages the complementary nature of thermal modality, which captures temperature

distributions to distinguish targets regardless of environmental lighting. Existing deep learning-based RGB-T SOD methods have various fusion strategies, categorized as early (Sun et al., 2019; Wang et al., 2020; Zhang et al., 2020a), intermediate (Wang et al., 2021; Zhou et al., 2021a), and late fusion (Han et al., 2017; Wang & Gong, 2019), to aggregate multimodal features. However, two significant bottlenecks remain. First, existing fixed fusion strategies often fail to adapt to varying environmental conditions. The reliability of RGB and thermal modalities shifts drastically across scenarios such as low illumination, complex backgrounds, or thermal noise, yet existing methods lack the mechanism to assess and reweight modality trustworthiness. Second, the high cost of manual pixel-level annotation for multimodal pairs results in relatively small-scale datasets. This data scarcity often leads to over-fitting, hindering robust feature learning and cross-scene generalization.

A promising direction to alleviate the issue of data scarcity is to leverage foundation models pre-trained on massive datasets, which possess powerful zero-shot generalization capabilities. Among these, the

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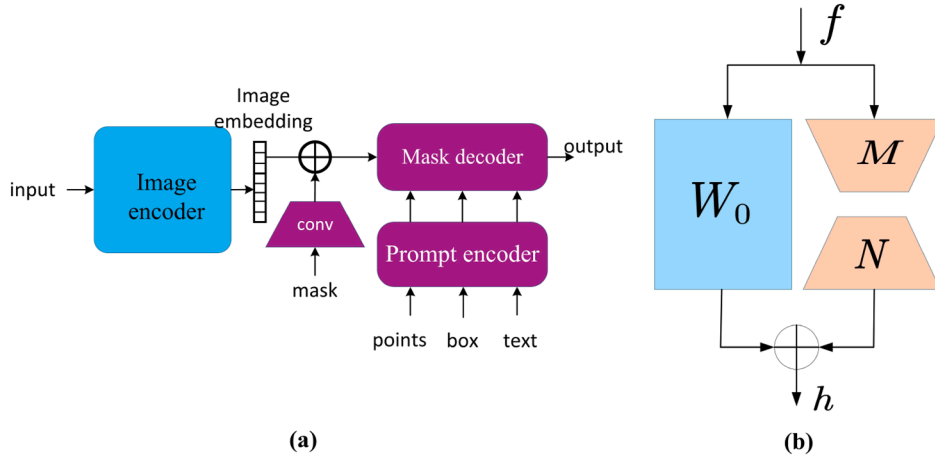


Fig. 1. (a) Schematic of the SAM framework. (b) LoRA.

Segment Anything Model (SAM) (Kirillov et al., 2023) stands out as a general-purpose segmentation foundation model with strong zero-shot generalization and universal feature representation capability. This versatility enables it to be widely used in various vision tasks (Chen et al., 2023b; Li et al., 2025; Wang et al., 2023; Zhang et al., 2024), which also provides a promising foundation for RGB-T SOD. However, direct application of SAM to RGB-T SOD still faces significant challenges: (1) As a class-agnostic segmentation model, SAM lacks inherent saliency semantics to identify salient objects in complex scenes. (2) SAM is trained primarily on RGB images, making it less effective at processing the thermal images and modeling cross-modal relationships between RGB and thermal modalities. (3) SAM relies on manual prompts (such as points or bounding boxes), which precludes the possibility of fully automated reasoning. Therefore, the key challenge we face is to effectively exploit SAM's extensive pre-trained knowledge and strong generalization while simultaneously equipping it with cross-modal saliency-awareness and prompt-free automation for RGB-T SOD.

To solve the above problems, we propose a Multi-modal Interaction Enhanced Segment Anything Model (MIE-SAM) to specialize SAM for multi-modal RGB-T saliency awareness. MIE-SAM adapts SAM's pre-trained knowledge to cross-modal salient object tasks with minimal parameter overhead. Specifically, we re-purpose SAM's original single-stream image encoder into a weight-sharing dual-branch structure to independently process RGB and thermal modalities. To inject saliency-specific semantics and bridge the modality gap without disrupting SAM's foundational weights, we present a Multi-modal Low-Rank Adaptation module (Mm-LoRA) nested within the frozen backbone layers. This allows for parameter-efficient fine-tuning while facilitating deep interactive fusion between RGB and thermal features. Second, to address the inflexibility of fixed fusion strategies, a Dynamic Fusion Module (DFM) is introduced to learn dynamic fusion weights and adaptively aggregate the multi-modal embeddings based on their environmental reliability. This ensures that the decoder receives robust, context-aware fused features. Finally, to eliminate the need for manual prompts and the complex modification of SAM's original prompt encoder and mask decoder, a Progressive Decoding Module (PDM) is designed to directly translate the fused features into fine-grained saliency masks in an entirely prompt-free, end-to-end manner. The main contributions of our work can be summarized as follows:

- We propose a novel RGB-T SOD framework MIE-SAM that extends the single-modal SAM encoder into a weight-sharing dual-branch system. The dual-modality feature representations achieve interactive fusion through multi-modal low-rank adaptation, and further fuse through adaptive dynamic weights to guide the decoder for gradually generating saliency masks. This design effectively preserves SAM's pre-trained

knowledge while instilling specialized multi-modal saliency awareness, enabling robust performance under complex environmental conditions.

- We develop the Multi-modal Low-Rank Adaptation (Mm-LoRA) module. Distinct from standard LoRA variants that tune single-modal features, Mm-LoRA serves a dual purpose: it injects task-specific knowledge into the frozen SAM backbone while simultaneously driving intra-layer cross-modal interaction at a low parameter cost.

- We design the Dynamic Fusion Module (DFM) to adaptively integrate RGB and thermal embeddings via learned weights, ensuring robust integration of complementary features. By dynamically fusing features based on the contribution of each modality, DFM ensures that the decoder receives environmentally adaptive fused features, improving performance in heterogeneous environments.

- We conduct extensive experiments to verify the robustness and generalization of the proposed MIE-SAM on three RGB-T SOD datasets.

2. Related work

2.1. RGB-T salient object detection

RGB-T SOD, as a multi-modal task that fuses visible and thermal infrared images, has received extensive attention in recent years. This task aims to overcome the limitations of RGB single modality in low light, occlusion, and adverse weather scenarios by combining complementary information from thermal images, thereby improving the robustness and accuracy of salient object detection.

In unsupervised methods (Tu et al., 2019; Wang et al., 2018; Yuan et al., 2024; Zhou et al., 2021b), Wang et al. (2018) proposed a multi-task manifold ranking algorithm based on the corresponding modal reliability and the cross-modality consistent constraint with the RGB and thermal features. Tu et al. (2019) proposed a collaborative graph learning method, in which superpixels are used as graph nodes and hierarchical deep features are employed to learn graph affinity and node saliency. These models rely heavily on heuristic hand-crafted features, which can result in limited generalization ability in complex scenes.

With the rapid development of deep learning, Convolutional Neural Network (CNN) based RGBT SOD (Bi et al., 2023; Tu et al., 2021; Xu et al., 2022; Zhang et al., 2019) has become the mainstream. Zhang et al. (2019) proposed an end-to-end RGB-T SOD model based on CNN feature fusion, which consists of three components: multiple adjacent-depth feature combination (ADFC) module, multi-branch group fusion (MGF) module, and a joint attention guided bi-directional message passing (JABMP) module. Tu et al. (2021) proposed a multi-interaction dual decoder network by deeply fusing multi-modal features, multi-level features, and global context at the decoding stage. With the introduction

of Transformer, multi-modal deep learning networks (Jiang et al., 2024; Liu et al., 2021; Lv et al., 2024; Pang et al., 2023; Zhang et al., 2023) have been further enriched. For example, Pang et al. (2023) proposed a Cross Modal View Hybrid Transformer (CAVER) model, which cascades several cross modal integrated units to realize a top-down information propagation path based on transformers. Liu et al. (2021) proposed a cross-modality fusion model SwinNet, which effectively fuses RGB-T data through spatial alignment and channel recalibration module and an edge-guided decoder. Although these methods have made significant progress, there is still room for improvement in their generalization ability due to the limited training data scale.

2.2. Application of SAM

With its powerful feature extraction and generalization ability accumulated by massive data pre-training, SAM (Kirillov et al., 2023) has shown a wide range of application potential in many fields of computer vision.

Some studies use SAM segmentation results as a prior to assist downstream tasks (Chen et al., 2023a; Han et al., 2023; Song et al., 2024; Wang et al., 2024c). For example, Han et al. (2023) proposed the open-vocabulary detector Sambor, which introduces the SideFormer module for acquiring SAM features and devises an Open-set RPN for leveraging SAM proposals to find potential objects. Based on pseudo labels of specific classes (such as CAM), Chen et al. (2023a) select the most relevant mask from the masks generated by SAM as the high-quality pseudo label to support weakly supervised and unsupervised image segmentation tasks.

Another research direction improves the performance of SAM in downstream tasks by injecting knowledge of downstream tasks into SAM through fine-tuning techniques (Cui et al., 2023; Peng et al., 2024; Wu et al., 2025b,c). Cui et al. (2023) fine-tune the pre-trained SAM model with the AdaLoRA strategy for segmenting salient object, thereby improving its performance on the RGB SOD task. Wu et al. (2025b) propose the Medical SAM Adapter, which adapts the pre-trained SAM to medical image segmentation via a parameter-efficient adapter-based fine-tuning strategy, significantly improving performance across diverse medical imaging modalities.

Furthermore, some work (Ma et al., 2025; Wang et al., 2024a; Wu et al., 2025a) studies the combination of SAM and multi-modal data to further expand the application scope. For example, Ma et al. (2025) proposed a unified multi-modal remote sensing semantic segmentation framework incorporating a novel multi-modal fine-tuning network (MFNet), which seamlessly integrate with Adapter and low-rank adaptation (LoRA). Wang et al. (2024a) proposed a novel framework called Sammese, which adapts the pre-trained SAM model to the RGB-T salient object detection task through multi-modal complementary fusion modules, multi-modal adapters, and semantically geometric cue generation. By combining the semantic prior knowledge of SAM into the fusion Net, Wu et al. (2025a) proposed the dual-layer optimization distillation mechanism which effectively transfers the SAM's semantic knowledge to the lightweight sub-network during training phase.

3. Method

3.1. Preliminary

Segment Anything Model: The architecture of SAM consists of three core components as shown in Fig. 1(a): (a) Image encoder: a large-scale vision transformer network for generating feature representations of images; (b) Prompt encoder: lightweight encoder that encodes geometric prompts such as points, boxes and masks into prompt embeddings. (c) Mask decoder: a lightweight transformer decoder that integrates image embeddings and prompt embeddings to output a predicted target mask. By training on very large-scale datasets, SAM obtains strong visual prior knowledge and excellent zero-shot generalization performance.

LoRA (Low Rank adaptation): LoRA (Low Rank adaptation) is an efficient technique for model fine-tuning. It maintains most of the parameter frozen and learns task-specific information with very few parameters. This high parameter efficiency learning enables the successful application of large foundation models to a wider range of downstream tasks. The core idea is to freeze the original weight matrix and represent the weight update part as the product of two low-rank matrices, which can adapt to new data while maintaining fewer training parameters. As shown in Fig. 1(b), for the pretrained weight matrix $W_0 \in \mathbb{R}^{c \times \hat{c}}$ and the input $f \in \mathbb{R}^{N \times c}$, LoRA introduces two low-rank matrices $M \in \mathbb{R}^{c \times r}$ and $N \in \mathbb{R}^{r \times \hat{c}}$ (where $r \ll \min(c, \hat{c})$) to indicate the weight update ΔW , the forward propagation calculation adjusted by LoRA is formulated as:

$$h = f(W_0 + \Delta W) = fW_0 + f\Delta W = fW_0 + fMN. \quad (1)$$

3.2. MIE-SAM

The architecture of the proposed MIE-SAM method is depicted in Fig. 2. MIE-SAM comprises three essential parts: the weight-sharing dual image encoders equipped with Multi-modal Low-Rank Adaptation (Mm-LoRA) for modality-specific perception and inter-modal interaction, Dynamic Fusion Module (DFM) for adaptively weighted integration of multi-modal features, and Progressive Decoder Module (PDM) for high-resolution saliency inference. Specifically, given RGB image and thermal image pair (I_{rgb}, I_t) , the weight-sharing dual SAM image encoders encode and interact the two modalities via Single-modal Low-Rank Adaptation (Sm-LoRA) and Cross-modal Low-Rank Adaptation (Cm-LoRA), resulting in the multi-modal embeddings $(\hat{F}_{rgb}, \hat{F}_t)$. These features are then aggregated by the DFM, which learns dynamic, environment-aware fusion weights to produce the fused embedding F_{fused} . Finally, the PDM takes the fused output F_{fused} as input and performs progressively upsampling the fused representation to generate the saliency map S without requiring any manual prompts. The following sections are devoted to the specific details of these components.

3.2.1. Multi-modal saliency feature extraction via multi-modal low-rank adaptation (Mmm-LoRA)

The general knowledge of SAM primarily resides in its image encoder. To extend SAM to multimodal scenarios while preserving this pre-trained knowledge, we require a fine-tuning strategy that is both parameter-efficient and modality interaction. Following the multimodal low-rank adaptation framework proposed in Ma et al. (2025), which has demonstrated superior performance in remote sensing, we integrate this architecture to bridge the gap between RGB and thermal modalities. The Multi-modal Low-Rank Adaptation (Mm-LoRA) is introduced for deeper cooperation between modality-specific learning and cross-modal interaction through efficient parameter fine-tuning in a multimodal interaction manner. We reconfigure the image encoder into a weight-sharing dual-branch image encoders, assigning one branch to each modality. RGB-T SOD requires a deeper synergy between "modality-specific extraction" and "cross-modal interaction". Therefore, we employ the Mm-LoRA module, which refines the interaction mechanism through two decoupled components: Single-modal LoRA (Sm-LoRA) for learning modality-specific representations, and Cross-modal LoRA (Cm-LoRA) for facilitating information exchange between branches, as illustrated in Fig. 2. Sm-LoRA focuses on adapting the weight-sharing encoder to the unique characteristics of each modality, ensuring that saliency features in thermal images and RGB images are accurately captured. Cm-LoRA module serves as a bridge between the two branches, enabling the interaction of complementary information. By exchanging low-rank feature residues, Cm-LoRA allows the model to leverage thermal data to compensate for RGB degradation in low-light or cluttered environments. This decomposition is formulated as follows:

Modality-specific Learning via Sm-LoRA: To effectively capture modality-specific salient features, the standard LoRA modules are integrated with the attention layers of their respective branches, facilitating independent feature learning for each modality. As shown

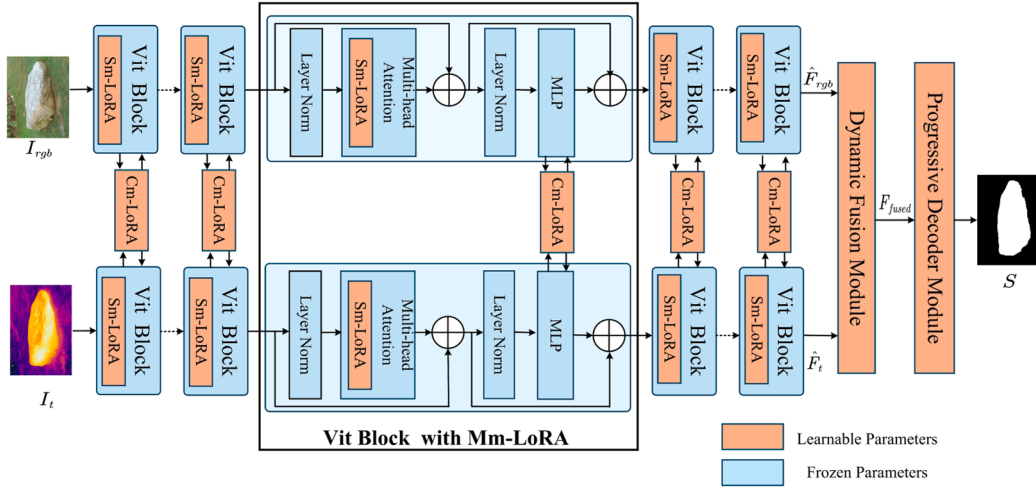


Fig. 2. Overview of the proposed MIE-SAM. It consists of dual-branch SAM's image encoders with Mm-LoRA (Multi-modal Low-Rank Adaptation), DFM (Dynamic Fusion Module) and PDM (Progressive Decoder Module).

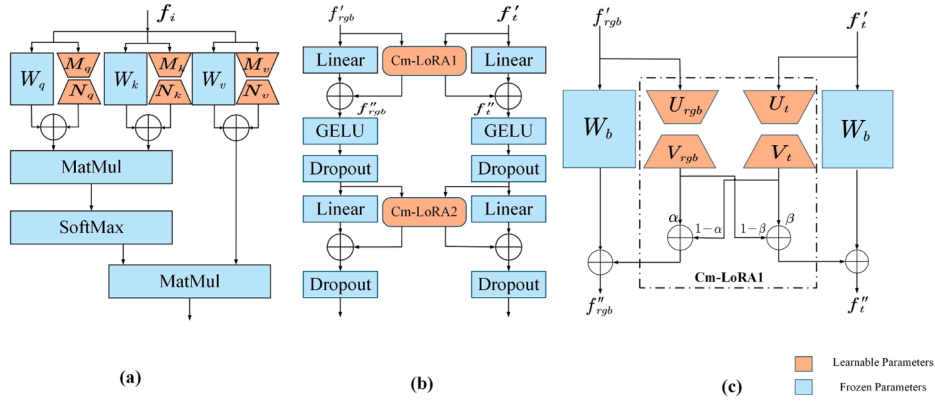


Fig. 3. (a) Sm-LoRA in the attention layer. (b) Cm-LoRA in the MLP layer. (c) The detailed structure of the Cm-LoRA1.

in Fig. 3(a), given one of the multi-modal input representations $f_i \in \mathbb{R}^{N \times c}$ ($i \in \{rgb, t\}$), the LoRA in the multi-head attention layer can be expressed as follows:

$$Q_{lora}^i = (W_q + N_q^i M_q^i) f_i$$

$$K_{lora}^i = (W_k + N_k^i M_k^i) f_i$$

$$V_{lora}^i = (W_v + N_v^i M_v^i) f_i$$

$$MHSA_{lora}^i(f_i) = \left(\frac{Q_{lora}^i K_{lora}^i}{\sqrt{k}} \right)^T V_{lora}^i$$

The original projection matrices $W_q \in \mathbb{R}^{c \times \hat{c}}$, $W_k \in \mathbb{R}^{c \times \hat{c}}$ and $W_v \in \mathbb{R}^{c \times \hat{c}}$ are shared across both branches and kept frozen, while $N_q^i \in \mathbb{R}^{c \times r}$, $M_q^i \in \mathbb{R}^{r \times \hat{c}}$, $N_k^i \in \mathbb{R}^{c \times r}$, $M_k^i \in \mathbb{R}^{r \times \hat{c}}$, $N_v^i \in \mathbb{R}^{c \times r}$ and $M_v^i \in \mathbb{R}^{r \times \hat{c}}$ ($i \in \{rgb, t\}$) are trainable parameters specific to the dual-branches, thereby enabling independent learning of modality-specific features.

Cross-modal Interaction via Cm-LoRA: To enrich and supplement modality-specific characteristics by leveraging cross-modal complementary information, we introduce the Cross-modal Low-Rank Adaptation (Cm-LoRA) within the MLP layers to facilitate effective cross-modal information fusion. As shown in Fig. 3(b), both fully-connected layers in the MLP incorporate the Cm-LoRA modules for promoting cross-modal interaction. Specifically, as depicted in Fig. 3(c), the first Cm-LoRA (Cm-LoRA1) embedded in the first fully connected layer realizes cross-modal interaction through residual summation with the other modality by weighting fusion parameters α, β . α and β are learnable parameters that control the contribution of cross-modal information. The Cm-LoRA1 in

the first fully connected layer for multi-modal input features f'_{rgb} and f'_t can be expressed as:

$$\begin{aligned} f''_{rgb} &= W_b(f'_{rgb}) + \alpha U_{rgb} V_{rgb}(f'_{rgb}) + (1 - \alpha) U_t V_t(f'_t) \\ f''_t &= W_b(f'_t) + \beta U_t V_t(f'_t) + (1 - \beta) U_{rgb} V_{rgb}(f'_{rgb}) \end{aligned} \quad (4)$$

where $W_b \in \mathbb{R}^{c \times \hat{c}}$ is shared and frozen, $U_i \in \mathbb{R}^{c \times r}$, $V_i \in \mathbb{R}^{r \times \hat{c}}$ ($i \in \{rgb, t\}$) are learnable parameters specific to the dual-branches.

Then, similar to the Cm-LoRA1 in the first fully connected layer, the second fully connected layer uses Cm-LoRA2 to perform deep multi-modal interaction at a higher level of feature learning.

During the training phase, only the newly introduced low-rank factorization matrices and fusion scalars are optimized, while the backbone parameters of the SAM image encoders remain frozen.

3.2.2. Dynamic integration of multi-modal saliency feature via dynamic fusion module (DFM)

In RGB-T salient object detection, the reliability of individual modalities often varies with environmental conditions. Fixed fusion strategies may introduce noise when one modality is unreliable. Therefore, we design a Dynamic Fusion Module (DFM), which computes input-dependent weights to adaptively balance the contributions of RGB and thermal modalities. As shown in Fig. 4, a weight filter, consisting of two convolutional layers with 3×3 kernels, a GELU activation function, and a softmax layer, processes the joint representation $\hat{F} = \text{Concat}(\hat{F}_{rgb}, \hat{F}_t)$ into the modality-aware weight $\lambda \in \mathbb{R}^{B \times N}$. By leveraging the modality-aware weight $\lambda \in \mathbb{R}^{B \times N}$ to compute a weighted average of $\hat{F}_{rgb}$ and $\hat{F}_t$,

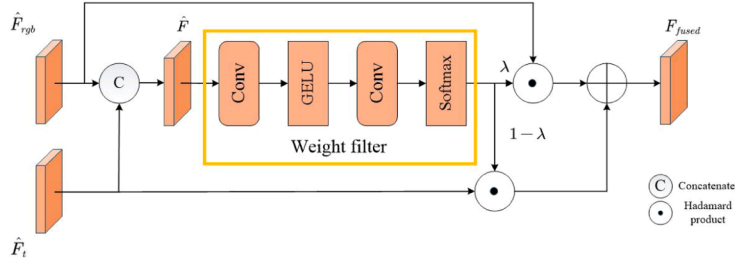


Fig. 4. Dynamic fusion module.

we obtain a fused multi-modal representation F_{fused} , which allows the model to prioritize the more informative modality under challenging conditions.

$$\lambda = softmax(\phi_{3 \times 3}(Ge(\phi_{3 \times 3}(\hat{F})))) \quad (5)$$

$$F_{fused} = \lambda \hat{F}_{rgb} + (1 - \lambda) \hat{F}_t \quad (6)$$

where $\phi_{3 \times 3}$ denotes the convolutional layer with 3×3 kernels, Ge stands for the GELU activation function.

The final fusion output F_{fused} is sent to the progressive decoder module for predicting the final saliency map.

3.2.3. Saliency refinement via progressive decoder module (PDM)

The fused feature representation F_{fused} still has a relatively low spatial resolution due to the encoder architecture, which conflicts with the requirement for high-precision saliency masks. To gradually restore the fused semantic features F_{fused} to the original input resolution and generate accurate pixel-wise saliency predictions, we design a Progressive Decoder Module (PDM) to recover lost spatial details through a hierarchical upsampling scheme. By alternating between upsampling with bilinear interpolation and refinement via 3×3 convolutional layers, PDM incrementally restores the feature F_{fused} to the original input size. Finally, two additional 3×3 convolutional layers followed by one 1×1 convolutional layer are applied to obtain the final saliency result S :

$$F^{up} = CBR_{3 \times 3}(US_{\times 2}(CBR_{3 \times 3}(US_{\times 2}(F_{fused})))) \quad (7)$$

$$S = CBR_{1 \times 1}(CBR_{3 \times 3}(CBR_{3 \times 3}(F^{up}))) \quad (8)$$

where $US_{\times 2}$ is a $2 \times$ upsampling operation using bilinear interpolation, $CBR_{3 \times 3}$ contains a convolutional layer with 3×3 kernels, a batch normalization layer, and a ReLU function.

4. Experiments

4.1. Experimental setup

4.1.1. Datasets

To evaluate the proposed MIE-SAM, we conduct experiments on three publicly available benchmark datasets: (1) The early benchmark dataset VT821 (Wang et al., 2018) is composed of 821 manually aligned image pairs, some of which are added with noise to increase detection difficulty. (2) VT1000 (Tu et al., 2019) includes 1000 RGB and thermal image pairs captured by a carefully aligned camera system, containing 400 objects in 10 simple scenes. (3) VT5000 (Tu et al., 2022b) contains 5000 pairs of spatially aligned RGB images and thermal imaging images, covering 11 challenging scenarios such as low light, occlusion, complex backgrounds to comprehensively evaluate the robustness of the algorithm under various conditions. To ensure fairness, we follow the setting of the existing research (Pang et al., 2023; Tu et al., 2021), using 2500 image pairs in the VT5000 dataset for training, and the remaining image pairs for testing.

4.1.2. Evaluation metrics

Precision-Recall Curve: The precision-recall (PR) curve is obtained by converting the predicted saliency map into a series of binary maps

using thresholds ranging from 0 to 255. Each binary map is then compared with the corresponding ground-truth mask to compute precision and recall values:

- Precision measures the proportion of correctly predicted foreground pixels among all pixels classified as salient.
- Recall denotes the proportion of ground-truth foreground pixels that are successfully detected.

MAE: The mean absolute error (MAE) (Borji et al., 2015) is the average absolute errors of all pixels between the predicted image P and the ground truth G :

$$MAE = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |P(i, j) - G(i, j)| \quad (9)$$

where H and W respectively represent the height and width of the saliency map.

F-Measure: F-Measure (Borji et al., 2015) is computed to provide a comprehensive evaluation of the model's performance by the weighted harmonic mean of Precision and Recall:

$$F_{\beta} = \frac{(1 + \beta^2) \times Precision \times Recall}{\beta^2 \times Precision + Recall} \quad (10)$$

where $\beta^2 = 0.3$.

S-Measure: S-Measure (Fan et al., 2017) aims to evaluate the structural similarity between the predicted result and the groundtruth. This metric is a linear combination of region structural similarity S_r and object-aware structural similarity S_o :

$$S_{\alpha} = \alpha \cdot S_o + (1 - \alpha) \cdot S_r \quad (11)$$

where α is usually set to 0.5.

E-Measure: The E-Measure (Fan et al., 2018) is an evaluation metric based on enhanced alignment mapping between the saliency map and the groundtruth:

$$E_{\xi} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \xi(i, j) \quad (12)$$

where ξ denotes an enhanced alignment matrix.

4.1.3. Experimental details

Our proposed framework MIE-SAM is implemented on an NVIDIA RTX 4090 GPU. The backbone is instantiated with the pretrained image encoder ViT-H in SAM. The input image is resized to 256×256 . During training, we adopt the (Adam et al., 2014) with the learning rate of 0.001, the batch size of 5 and the train epoch of 100 for optimization.

4.2. Comparisons to SOTAs

4.2.1. Quantitative comparisons

We compare the proposed MIE-SAM with 11 SOTA RGB-T SOD methods, including CNN based methods: MIDD (Tu et al., 2021), DCNet (Tu et al., 2022a), TNet (Cong et al., 2022), OSRNet (Huo et al., 2022), E2Net (Bi et al., 2023), LSNet (Zhou et al., 2023b), MCFNet (Ma et al., 2023), PCNet (Wu et al., 2024), LAFB (Wang et al., 2024b); Transformer-based methods: SwinNet (Liu et al., 2021), CGFNet (Wang et al., 2021),

Table 1

Quantitative comparisons with 11 state-of-the-art RGB-T SOD methods on three representative datasets in terms of E-m(E-measure), F-m(F-measure), S-m(S-measure) and MAE(Mean Absolute Error). $\uparrow$ and $\downarrow$ indicate larger and smaller is better respectively. The best results are shown in bold.

Methods	VT821				VT1000				VT5000			
	E-m $\uparrow$	S-m $\uparrow$	F-m $\uparrow$	MAE $\downarrow$	E-m $\uparrow$	S-m $\uparrow$	F-m $\uparrow$	MAE $\downarrow$	E-m $\uparrow$	S-m $\uparrow$	F-m $\uparrow$	MAE $\downarrow$
MIDD (Tu et al., 2021)	0.895	0.871	0.804	0.045	0.933	0.915	0.882	0.027	0.897	0.868	0.801	0.043
DCNet (Tu et al., 2022a)	0.912	0.876	0.823	0.033	0.948	0.922	0.902	0.021	0.920	0.871	0.819	0.035
TNet (Cong et al., 2022)	0.919	0.899	0.842	0.030	0.937	0.929	0.889	0.021	0.927	0.895	0.846	0.033
OSRNet (Huo et al., 2022)	0.896	0.875	0.813	0.043	0.935	0.926	0.892	0.022	0.908	0.875	0.823	0.040
E2Net (Bi et al., 2023)	0.916	0.880	0.847	0.035	0.943	0.908	0.891	0.027	0.938	0.897	0.826	0.031
LSNet (Zhou et al., 2023b)	0.911	0.878	0.825	0.033	0.935	0.925	0.885	0.023	0.915	0.877	0.825	0.037
MCFNet (Ma et al., 2023)	0.918	0.891	0.844	0.029	0.944	0.932	0.902	0.019	0.924	0.888	0.848	0.033
PCNet (Wu et al., 2024)	0.900	0.874	0.818	0.036	0.948	0.925	0.886	0.021	0.919	0.875	0.828	0.036
LAFB (Wang et al., 2024b)	0.915	0.884	0.842	0.034	0.945	0.932	0.905	0.018	0.931	0.893	0.857	0.030
SwinNet (Liu et al., 2021)	0.926	0.904	0.847	0.030	0.947	0.938	0.896	0.018	0.942	0.912	0.865	0.026
CGFNet (Wang et al., 2021)	0.912	0.881	0.829	0.038	0.944	0.923	0.900	0.023	0.922	0.883	0.831	0.035
CAVER (Pang et al., 2023)	0.919	0.891	0.839	0.033	0.945	0.936	0.903	0.017	0.924	0.892	0.842	0.032
WaveNet (Zhou et al., 2023a)	0.929	0.912	0.857	0.024	0.952	0.945	0.909	0.015	0.940	0.912	0.867	0.026
MIE-SAM	0.939	0.915	0.887	0.021	0.953	0.945	0.911	0.017	0.951	0.907	0.894	0.024

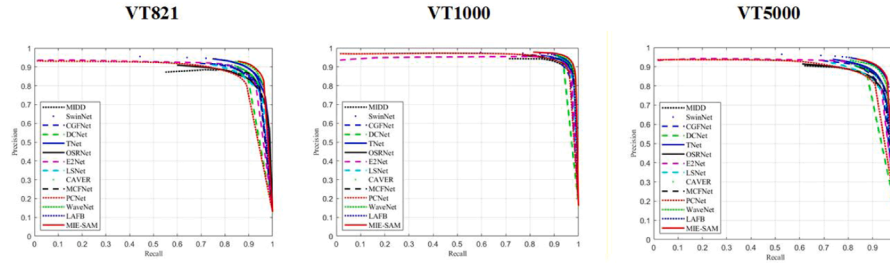


Fig. 5. P-R curves comparison with the state-of-the-art RGB-T SOD methods on VT5000, VT1000, and VT821 datasets.

CAVER (Pang et al., 2023); and Wavelet transform: WaveNet (Zhou et al., 2023a). We use the experimental results directly provided by the authors or re-run the provided code with default parameter settings for comparisons.

The quantitative results for the three RGB-T SOD datasets are shown in Table 1. The experimental results show that our proposed MIE-SAM exhibits significant advantages over these advanced methods in all four evaluation metrics. Specifically, MIE-SAM achieves the highest scores in all performance of E-m, S-m and F-m, surpassing WaveNet (Zhou et al., 2023a) with an average advantage of 0.6%, 0.3% and 1.9% on the VT821 and VT1000 datasets. For the VT5000 dataset, MIE-SAM performs best in E-m, F-m and MAE, improving by 1.0%, 3.1% and 7.7% respectively, compared with WaveNet (Zhou et al., 2023a). Furthermore, its MAE indicator decreases by an average of 10% compared with WaveNet (Zhou et al., 2023a) on VT821 and VT5000. In Fig. 5, the precise-recall curves on the three RGB-T SOD datasets show that our method achieves the highest accuracy on all three datasets. This consistent performance advantage across different datasets and evaluation criteria strongly validates the effectiveness of the MIE-SAM framework for RGB-T SOD tasks, while demonstrating its strong generalization ability.

4.2.2. Qualitative comparison

The qualitative comparison in Fig. 6 show the superiority of MIE-SAM in different scenarios. For example, in rows 1–3, the temperature of the salient object is similar to the background; in rows 4–6, salient objects are located in scenes with complex backgrounds and cluttered textures; in rows 7–9, multiple salient objects are disturbed by background temperature noise; in rows 10–12, the object is in a low light environment. For these challenging situations, most existing models fail to fully detect the entire salient object region due to insufficient fusion methods while MIE-SAM consistently generates saliency maps with finer structures, more complete regions, and clearer edges. These visual

results show that by effectively utilizing the proposed cross-modal interaction mechanism in Mm-LoRA and DFM, MIE-SAM achieves more visually satisfactory segmentation quality in different scene types.

4.2.3. Challenge-sensitive performance

To assess the robustness of our method under challenging conditions, we conduct attribute-based evaluations across 13 annotated scenarios of the VT5000 dataset. Table 2 reports the F-measure comparisons between MIE-SAM and six representative state-of-the-art RGB-T SOD methods. The results demonstrate that MIE-SAM consistently achieves the best performance on 12 out of 13 challenging attributes, confirming the overall superiority of our framework in handling diverse scene variations. Although its performance in the "bad weather" attribute is slightly lower than that of CAVER (Pang et al., 2023), MIE-SAM still maintains a highly competitive result. This outstanding performance under various conditions highlights the effectiveness of our framework in fusing complementary RGB-T information and adapting to diverse scene variations.

Extending this evaluation to the VT821 dataset (Table 3), MIE-SAM again achieves the highest F-measure on 9 of 11 attributes and stays highly competitive on the remaining two. These improvements are particularly evident in challenging scenarios such as background clutter, multi-object scenes and thermal confusion. We attribute this success to two key components: the Mm-LoRA module for adaptive cross-modal interaction and the DFM for reliability-guided fusion, which collectively reduce modality-specific noise and improve the distinguishability of salient features. These cross-dataset results demonstrate that MIE-SAM is robust and generalizable across diverse real-world conditions.

Fig. 7 presents qualitative comparisons between MIE-SAM and several representative RGB-T SOD methods under various challenging scenarios from VT5000 and VT821, including complex backgrounds, low illumination, thermal confusion and small targets. As observed, existing methods tend to produce incomplete object masks, inaccurate bound-

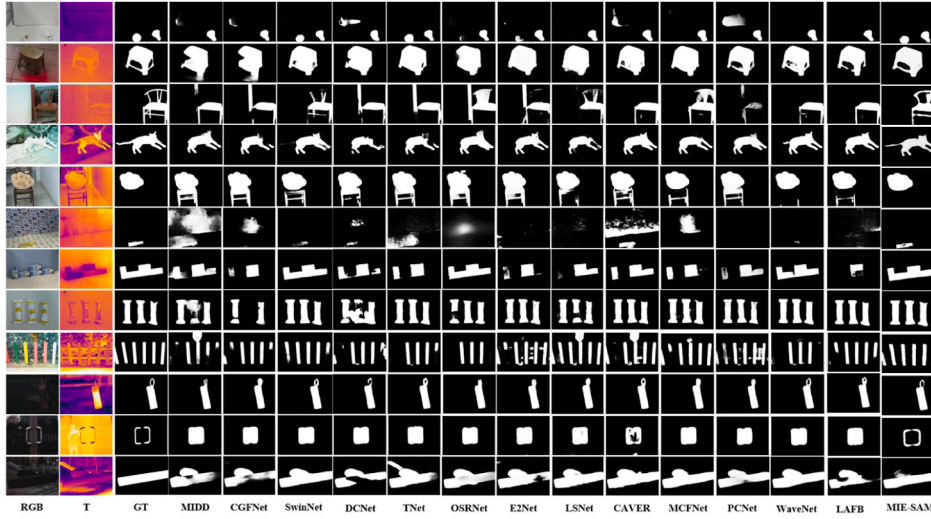


Fig. 6. Visualization comparison of MIE-SAM with the state-of-the-art methods.

Table 2

Performance comparison (F-measure) on 13 challenging attributes of the VT5000.

Attributes	BSO	CB	CIB	IC	LI	MSO	OF	SSO	SA	TC	BW	RGB	T
TNet (Cong et al., 2022)	0.845	0.835	0.816	0.793	0.844	0.822	0.822	0.799	0.825	0.819	0.767	0.844	0.845
OSRNet (Huo et al., 2022)	0.838	0.801	0.798	0.764	0.819	0.782	0.793	0.707	0.764	0.769	0.766	0.810	0.815
MCFNet (Ma et al., 2023)	0.858	0.827	0.839	0.788	0.858	0.805	0.820	0.762	0.813	0.806	0.804	0.840	0.842
SwinNet (Liu et al., 2021)	0.879	0.835	0.820	0.820	0.870	0.819	0.843	0.738	0.825	0.827	0.786	0.846	0.850
CGFNet (Wang et al., 2021)	0.837	0.825	0.808	0.792	0.834	0.786	0.816	0.768	0.808	0.806	0.750	0.836	0.837
CAVER (Pang et al., 2023)	0.870	0.844	0.858	0.810	0.863	0.817	0.825	0.780	0.815	0.827	0.813	0.854	0.853
MIE-SAM	0.899	0.883	0.879	0.856	0.889	0.875	0.859	0.816	0.881	0.887	0.805	0.896	0.894

Table 3

Performance comparison (F-measure) on 11 challenging attributes of the VT821.

Attributes	BSO	CB	CIB	IC	LI	MSO	OF	SSO	SA	TC	BW
TNet (Cong et al., 2022)	0.843	0.787	0.827	0.825	0.853	0.832	0.784	0.699	0.846	0.837	0.812
OSRNet (Huo et al., 2022)	0.843	0.753	0.826	0.783	0.807	0.810	0.763	0.654	0.801	0.799	0.832
MCFNet (Ma et al., 2023)	0.856	0.770	0.831	0.823	0.842	0.836	0.786	0.686	0.844	0.833	0.839
SwinNet (Liu et al., 2021)	0.863	0.776	0.859	0.847	0.871	0.850	0.802	0.687	0.853	0.858	0.875
CGFNet (Wang et al., 2021)	0.845	0.788	0.831	0.828	0.861	0.829	0.801	0.724	0.840	0.838	0.829
CAVER (Pang et al., 2023)	0.869	0.777	0.848	0.834	0.841	0.836	0.800	0.685	0.816	0.841	0.856
MIE-SAM	0.891	0.809	0.883	0.862	0.866	0.882	0.815	0.725	0.867	0.871	0.859

aries, or false positives in these cases. In contrast, MIE-SAM generates saliency maps with more complete coverage, clearer edges, and reduced background noise. These visual results are consistent with the quantitative findings, further confirming the advantage of our method in handling challenging scenes.

In summary, the results from attribute-based quantitative evaluations and the qualitative visual comparisons on VT5000 and VT821 collectively demonstrate that MIE-SAM consistently outperforms existing state-of-the-art methods under diverse and challenging conditions. This performance advantage is primarily attributed to the proposed cross-modal interaction module and the adaptive fusion strategy, which enable the model to effectively integrate complementary information from RGB and thermal modalities while dynamically prioritizing the more reliable source according to environmental conditions. These findings suggest that incorporating dynamic multi-modal reasoning into foundation models such as SAM offers a viable direction for improving robustness and generalization in multi-modal vision tasks.

4.3. Ablation study

To evaluate the contribution of each core component in the proposed MIE-SAM, we conducted a series of ablation experiments on the

Table 4

Quantitative results for different fine-tuning mechanisms on the VT5000 dataset. The best results are shown in bold.

Modality	Fine-tuning	DFM	PDM	E-m↑	S-m↑	F-m↑	MAE↓
RGB+T	×	×	✓	0.902	0.835	0.802	0.047
RGB+T	LoRA	×	✓	0.916	0.892	0.829	0.036
RGB+T	Mm-LoRA	×	✓	0.932	0.900	0.851	0.031
RGB+T	Mm-LoRA	✓	✓	0.951	0.907	0.894	0.024

VT5000 dataset. The analysis focuses on two primary modules: Multi-modal Low-Rank Adaptation (Mm-LoRA) in dual-branch image encoders and the Dynamic Fusion Module (DFM).

Mm-LoRA (Multi-modal Low-Rank Adaptation): We first evaluate the role of Mm-LoRA on multi-modal learning within the dual-branch image encoders. As shown in Table 4 (rows 1–3), the baseline model, which directly adopts the dual-branch SAM image encoders with the Progressive Decoder Module (PDM), yields a reasonable starting point for RGB-T salient object detection. Introducing the standard LoRA into the dual-branch image encoders improves the adaptation capability of the pretrained SAM encoder via parameter-efficient fine-tuning. However, independently optimizing each modality still limits cross-

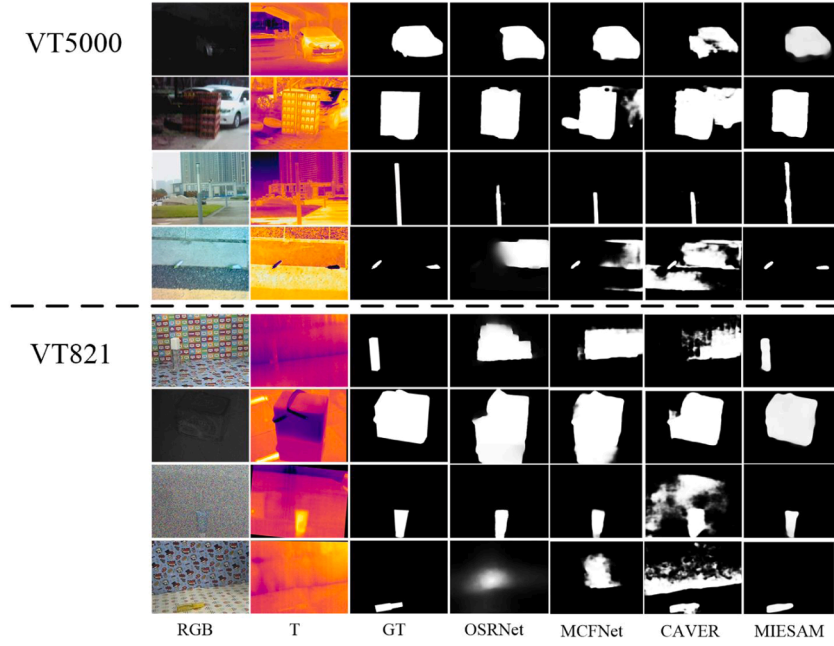


Fig. 7. Attribute-based visual comparison of MIE-SAM with the state-of-the-art methods.

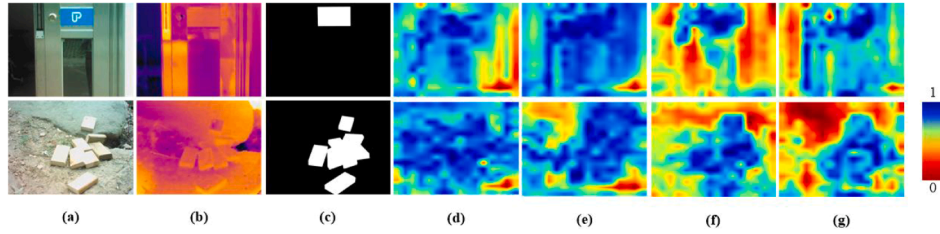


Fig. 8. (a) RGB image. (b) T image. (c) GT. (d) The RGB feature map of the RGB-T dual-branch image encoder using the standard LoRA. (e) The t feature map of the RGB-T dual-branch image encoder using the standard LoRA. (f) The RGB feature map of the RGB-T dual-branch image encoder using Mm-LoRA. (g) The t feature map of the RGB-T dual-branch image encoder using Mm-LoRA.

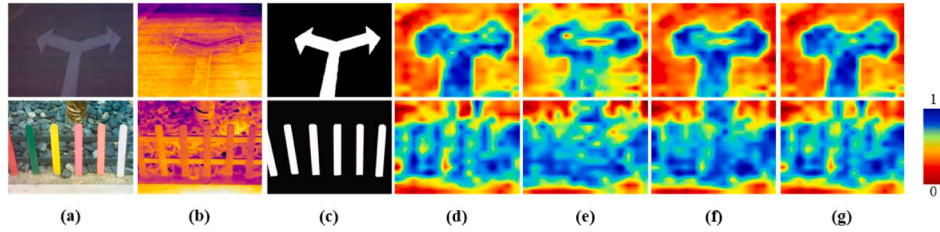


Fig. 9. (a) RGB image. (b) T image. (c) GT. (d) The RGB feature map of the RGB-T dual-branch image encoder using Mm-LoRA. (e) The T feature map of the RGB-T dual-branch image encoder using Mm-LoRA. (f) Feature fusion by simple summation. (g) Feature fusion by the proposed DFM. The learned modality weights are (RGB: 0.83, T: 0.17) for the first example and (RGB: 0.77, T: 0.23) for the second example, illustrating the adaptive reliability-aware fusion mechanism.

modal interaction, leading to insufficient utilization of complementary RGB-T information. In contrast, incorporating our proposed Mm-LoRA leads to substantial performance gains (row 3 vs. row 2), demonstrating its effectiveness in facilitating cross-modal interaction and feature fusion. This advantage is further confirmed by the visual results in Fig. 8: the feature maps generated by the standard LoRA from RGB and thermal inputs exhibit poor foreground-background discriminability, with substantial background regions being erroneously activated as foreground (Fig. 8(d)–(e)), whereas Mm-LoRA effectively suppresses such false responses and generates more structurally coherent feature representations that better preserve object boundaries in complex scenes (Fig. 8(f)–(g)).

DFM (Dynamic Fusion Module): We analyze the importance of Dynamic Fusion Module (DFM) in Table 4 (rows 3–4). Building upon

Table 5
Effects of different LoRA rank.

LoRA	E-meas↑	S-meas↑	F-meas↑	MAE↓
rank = 2	0.947	0.911	0.874	0.026
rank = 4	0.951	0.907	0.894	0.024
rank = 8	0.949	0.912	0.882	0.027
rank = 16	0.943	0.906	0.860	0.028

the dual-branch image encoders equipped with Mm-LoRA, DFM learns modality-aware dynamic weights and adaptively integrates RGB and thermal features, enabling the model to effectively exploit complementary information according to per-modal reliability. Quantitative results

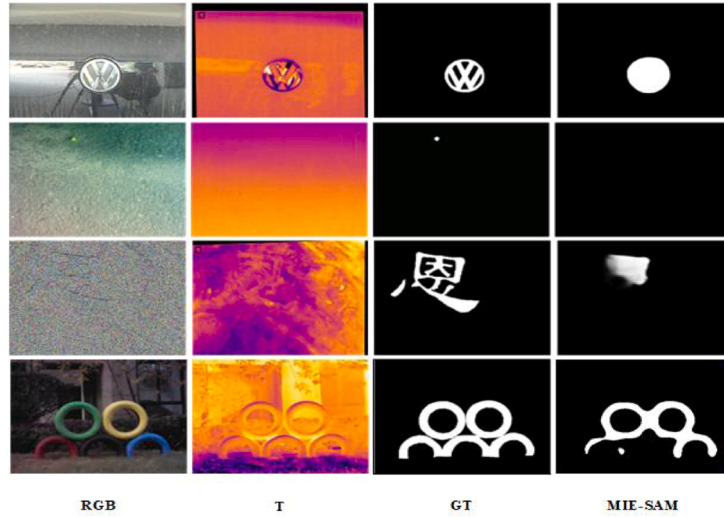


Fig. 10. Visual results of MIE-SAM in some typical failure cases.

Table 6
Comparison of the complexity of RGB-T SOD methods.

Method	Backbone	FLOPS(G)	Params.(M)	FPS
MIDD (Tu et al., 2021)	VGG-16	216.7	52.4	22
SwinNet (Liu et al., 2021)	Swin-B	124.3	198.7	10
CGFNet (Wang et al., 2021)	VGG-16	345.1	66.4	13
OSRNet (Huo et al., 2022)	VGG-16	34.6	15.6	53
CAVER (Pang et al., 2023)	ResNet-50d	44.4	55.8	27
MIESAM	ViT-H	217.9	11.6	3

in Table 4 show that incorporating DFM brings consistent performance gains over the Mm-LoRA baseline with sum fusion. This observation is further verified in Fig. 9. Compared with simple summation (Fig. 9(f)), the proposed DFM (Fig. 9(g)) generates cleaner fused representations with more complete object structures and reduced background interference. Furthermore, the learned fusion weights provide an intuitive interpretation of the adaptive fusion process. For the first example in row 1, the DFM assigns weights of 0.83 and 0.17 to the RGB and thermal modalities, respectively, indicating that the RGB modality provides the dominant visual cues under this scene. In the second example in row 2, although the RGB modality still contributes the largest proportion (0.77), the thermal modality weight increases to 0.23, demonstrating that DFM automatically enhances the contribution of thermal information as its reliability increases. These results confirm that the proposed reliability-aware dynamic weighting strategy can flexibly adjust the contribution of each modality according to the scene characteristics, rather than relying on a fixed fusion ratio, thereby producing more robust fused representations for RGB-T salient object detection.

Hyper-parameter Sensitivity Analysis: We analyze the sensitivity of the LoRA rank, which controls the adaptation capacity of the low-rank matrices. As shown in Table 5, the performance of our method remains relatively stable for ranks ranging from 2 to 8, indicating that MIE-SAM is low sensitivity to this hyperparameter within this range. Among the evaluated ranks, rank=4 achieves the best overall performance across E-measure, F-measure, and MAE. When the rank increases to 16, the performance drops noticeably, possibly due to overfitting or increased optimization difficulty with excessive trainable parameters. Therefore, rank=4 is selected as the default setting.

4.4. Complexity analysis

We evaluate model complexity in terms of trainable parameters, FLOPs, and inference speed (FPS), as reported in Table 6. Benefit-

ing from the proposed Mm-LoRA, our method introduces only 11.6M trainable parameters while keeping the pretrained SAM image encoder frozen—substantially fewer than those of existing RGB-T SOD methods. In terms of FLOPs and FPS, MIE-SAM adopts the ViT-H backbone, which incurs relatively high computational cost and lower inference speed. However, this overhead primarily stems from the frozen backbone rather than the lightweight Mm-LoRA and DFM modules. Given the accuracy gains achieved, this computational overhead is justifiable. Overall, these results demonstrate that our method achieves a favorable balance between detection performance and model complexity, highlighting the potential of parameter-efficient foundation-model adaptation for RGB-T SOD.

4.5. Discussion

Although MIE-SAM achieves superior performance on most challenging scenarios, it still exhibits limitations in certain extreme cases illustrated in Fig. 10. When salient objects present low contrast in thermal images and lack distinguishable appearance clues in RGB images (lines 1–2), both modalities provide ambiguous information, making it difficult for the model to reliably distinguish foreground from background. In scenes where salient objects present complex structures or local noises (lines 3–4), the global segmentation prior inherited from SAM can easily lead to false or missed detections of local details in salient objects. These failure cases show that the performance of MIE-SAM is still limited by the poor modal information quality in extreme scenarios and local noise interference in complex scenarios. Future work will explore finer-grained modal reliability estimation and local feature learning strategies to further improve robustness in such challenging scenarios.

5. Conclusion

In this paper, we present MIE-SAM (Multi-Modal Interaction Enhanced Segment Anything Model), a novel adaptation of the SAM architecture for RGB-T salient object detection. Our approach introduces three components: the Multi-modal Low-Rank Adaptive (Mm-LoRA) which enables parameter-efficient cross-modal knowledge transfer within dual-branch encoders, the Dynamic Fusion Module (DFM) which performs adaptive aggregation of RGB-T features via reliability-aware weighting and the Progressive Decoder Module (PDM) which refines fused representations into precise saliency predictions. This architectural design achieves salient object awareness in SAM while preserving its inherent generalization capability. Extensive experimental results

on public datasets demonstrate that MIE-SAM achieves superior performance, proving the enormous potential of utilizing foundational vision models for processing multi-modal data and performing multi-modal saliency detection tasks.

CRedit authorship contribution statement

Ze Li: Writing – original draft, Software, Methodology; **Ying Ying Zhang:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition; **Shuai Zhang:** Writing – review & editing, Software, Investigation; **Zhi Peng Wang:** Writing – review & editing, Software, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Adam, K. D. P. B. J. et al. (2014). A method for stochastic optimization, *1412*(6) arXiv:1412.6980.
- Bi, H., Zhang, J., Wu, R., Tong, Y., Fu, X., & Shao, K. (2023). Rgb-t salient object detection via excavating and enhancing cnn features. *Applied Intelligence*, *53*(21), 25543–25561.
- Borji, A., Cheng, M.-M., Jiang, H., & Li, J. (2015). Salient object detection: A benchmark. *IEEE Transactions on Image Processing*, *24*(12), 5706–5722.
- Chen, T., Mai, Z., Li, R., & Chao, W.-l. (2023a). Segment anything model (sam) enhanced pseudo labels for weakly supervised semantic segmentation. arXiv:2305.05803.
- Chen, T., Zhu, L., Deng, C., Cao, R., Wang, Y., Zhang, S., Li, Z., Sun, L., Zang, Y., & Mao, P. (2023b). Sam-adapter: Adapting segment anything in underperformed scenes. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 3367–3375).
- Cong, R., Zhang, K., Zhang, C., Zheng, F., Zhao, Y., Huang, Q., & Kwong, S. (2022). Does thermal really always matter for RGB-t salient object detection? *IEEE Transactions on Multimedia*, *25*, 6971–6982.
- Cui, R., He, S., & Qiu, S. (2023). Adaptive low rank adaptation of segment anything to salient object detection. arXiv:2308.05426.
- Fan, D.-P., Cheng, M.-M., Liu, Y., Li, T., & Borji, A. (2017). Structure-measure: A new way to evaluate foreground maps. In *Proceedings of the IEEE international conference on computer vision* (pp. 4548–4557).
- Fan, D.-P., Gong, C., Cao, Y., Ren, B., Cheng, M.-M., & Borji, A. (2018). Enhanced-alignment measure for binary foreground map evaluation. arXiv:1805.10421.
- Gao, Y., Shi, M., Tao, D., & Xu, C. (2015). Database saliency for fast image retrieval. *IEEE Transactions on Multimedia*, *17*(3), 359–369.
- Garcia-Garcia, A., Orts-Escolano, S., Oprea, S., Villena-Martinez, V., Martinez-Gonzalez, P., & Garcia-Rodriguez, J. (2018). A survey on deep learning techniques for image and video semantic segmentation. *Applied Soft Computing*, *70*, 41–65.
- Han, J., Chen, H., Liu, N., Yan, C., & Li, X. (2017). Cnns-based rgb-d saliency detection via cross-view transfer and multiview fusion. *IEEE Transactions on Cybernetics*, *48*(11), 3171–3183.
- Han, X., Wei, L., Yu, X., Dou, Z., He, X., Wang, K., Sun, Y., Han, Z., & Tian, Q. (2023). Boosting segment anything model towards open-vocabulary learning. arXiv:2312.03628.
- Hong, S., You, T., Kwak, S., & Han, B. (2015). Online tracking by learning discriminative saliency map with convolutional neural network. In *International conference on machine learning* (pp. 597–606). PMLR.
- Hoyer, L., Munoz, M., Katiyar, P., Khoreva, A., & Fischer, V. (2019). Grid saliency for context explanations of semantic segmentation. *Advances in Neural Information Processing Systems*, *32*.
- Huo, F., Zhu, X., Zhang, Q., Liu, Z., & Yu, W. (2022). Real-time one-stream semantic-guided refinement network for RGB-thermal salient object detection. *IEEE Transactions on Instrumentation and Measurement*, *71*, 1–12.
- Jiang, M., Ma, J., Chen, J., Wang, Y., & Fang, X. (2024). Patnet: Patch-to-pixel attention-aware transformer network for rgb-d and rgb-t salient object detection. *Knowledge-Based Systems*, *291*, 111597.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A. C., Lo, W.-Y. et al. (2023). Segment anything. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 4015–4026).
- Li, C., Liang, X., Lu, Y., Zhao, N., & Tang, J. (2019). Rgb-t object tracking: Benchmark and baseline. *Pattern Recognition*, *96*, 106977.
- Li, D., Xie, W., Cao, M., Wang, Y., Zhang, Y., Fang, L., Li, Y., & Xu, C. (2025). FusionSAM: Visual multi-modal learning with segment anything model. In *Proceedings of the 31st ACM SIGKDD conference on knowledge discovery and data mining v. 2* (pp. 1330–1340).
- Liu, J.-J., Hou, Q., Cheng, M.-M., Feng, J., & Jiang, J. (2019). A simple pooling-based design for real-time salient object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3917–3926).
- Liu, Z., Tan, Y., He, Q., & Xiao, Y. (2021). Swinnet: Swin transformer drives edge-aware RGB-d and RGB-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, *32*(7), 4486–4497.
- Lu, A., Li, C., Yan, Y., Tang, J., & Luo, B. (2021). Rgbt tracking via multi-adaptor network with hierarchical divergence loss. *IEEE Transactions on Image Processing*, *30*, 5613–5625.
- Lv, C., Zhou, X., Wan, B., Wang, S., Sun, Y., Zhang, J., & Yan, C. (2024). Transformer-based cross-modal integration network for RGB-t salient object detection. *IEEE Transactions on Consumer Electronics*, *70*(2), 4741–4755.
- Ma, S., Song, K., Dong, H., Tian, H., & Yan, Y. (2023). Modal complementary fusion network for RGB-t salient object detection. *Applied Intelligence*, *53*(8), 9038–9055.
- Ma, X., Zhang, X., Pun, M.-O., & Huang, B. (2025). A unified framework with multimodal fine-tuning for remote sensing semantic segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, *63*, 1–15. <https://doi.org/10.1109/TGRS.2025.3585238>
- Pang, Y., Zhao, X., Zhang, L., & Lu, H. (2023). Caver: Cross-modal view-mixed transformer for bi-modal salient object detection. *IEEE Transactions on Image Processing*, *32*, 892–904.
- Peng, Z., Xu, Z., Zeng, Z., Xie, L., Tian, Q., & Shen, W. (2024). Parameter efficient fine-tuning via cross block orchestration for segment anything model. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3743–3752).
- Shao, L., & Brady, M. (2006). Specific object retrieval based on salient regions. *Pattern Recognition*, *39*(10), 1932–1948.
- Song, Y., Zhou, Q., Lu, X., Shao, Z., & Ma, L. (2024). Su-sam: A simple unified framework for adapting segment anything model in underperformed scenes. arXiv:2401.17803.
- Sun, Y., Zuo, W., & Liu, M. (2019). Rtfnet: Rgb-thermal fusion network for semantic segmentation of urban scenes. *IEEE Robotics and Automation Letters*, *4*(3), 2576–2583.
- Tu, Z., Li, Z., Li, C., Lang, Y., & Tang, J. (2021). Multi-interactive dual-decoder for RGB-thermal salient object detection. *IEEE Transactions on Image Processing*, *30*, 5678–5691.
- Tu, Z., Li, Z., Li, C., & Tang, J. (2022a). Weakly alignment-free RGBT salient object detection with deep correlation network. *IEEE Transactions on Image Processing*, *31*, 3752–3764.
- Tu, Z., Ma, Y., Li, Z., Li, C., Xu, J., & Liu, Y. (2022b). Rgbt salient object detection: A large-scale dataset and benchmark. *IEEE Transactions on Multimedia*, *25*, 4163–4176.
- Tu, Z., Xia, T., Li, C., Wang, X., Ma, Y., & Tang, J. (2019). Rgb-t image saliency detection via collaborative graph learning. *IEEE Transactions on Multimedia*, *22*(1), 160–173.
- Wang, D., Zhang, J., Du, B., Xu, M., Liu, L., Tao, D., & Zhang, L. (2023). Samrs: Scaling-up remote sensing segmentation dataset with segment anything model. *Advances in Neural Information Processing Systems*, *36*, 8815–8827.
- Wang, G., Li, C., Ma, Y., Zheng, A., Tang, J., & Luo, B. (2018). Rgb-t saliency detection benchmark: Dataset, baselines, analysis and a novel approach. In *Chinese conference on image and graphics technologies* (pp. 359–369). Springer.
- Wang, J., Song, K., Bao, Y., Huang, L., & Yan, Y. (2021). Cgfn: Cross-guided fusion network for rgb-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, *32*(5), 2949–2961.
- Wang, K., Lin, D., Li, C., Tu, Z., & Luo, B. (2024a). Adapting segment anything model to multi-modal salient object detection with semantic feature fusion guidance. arXiv:2408.15063.
- Wang, K., Tu, Z., Li, C., Zhang, C., & Luo, B. (2024b). Learning adaptive fusion bank for multi-modal salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, *34*(8), 7344–7358.
- Wang, N., & Gong, X. (2019). Adaptive fusion for RGB-d salient object detection. *IEEE access*, *7*, 55277–55284.
- Wang, X., Li, S., Chen, C., Fang, Y., Hao, A., & Qin, H. (2020). Data-level recombination and lightweight fusion scheme for RGB-d salient object detection. *IEEE Transactions on Image Processing*, *30*, 458–471.
- Wang, X., Ye, F., & Zhang, Y. (2024c). Task-aware low-rank adaptation of segment anything model. arXiv:2403.10971.
- Wu, G., Liu, H., Fu, H., Peng, Y., Liu, J., Fan, X., & Liu, R. (2025a). Every SAM drop counts: Embracing semantic priors for multi-modality image fusion and beyond. In *Proceedings of the computer vision and pattern recognition conference* (pp. 17882–17891).
- Wu, J., Wang, Z., Hong, M., Ji, W., Fu, H., Xu, Y., Xu, M., & Jin, Y. (2025b). Medical sam adapter: Adapting segment anything model for medical image segmentation. *Medical Image Analysis*, *102*, 103547.
- Wu, R., Bi, H., Zhang, C., Zhang, J., Tong, Y., Jin, W., & Liu, Z. (2024). Pyramid contract-based network for RGB-t salient object detection. *Multimedia Tools and Applications*, *83*(7), 20805–20825.
- Wu, Y., Wang, Z., Yang, X., Kang, H., He, A., & Li, T. (2025c). Trans-sam: Transfer segment anything model to medical image segmentation with parameter-efficient fine-tuning. *Knowledge-Based Systems*, *310*, 112909.
- Wu, Y.-H., Gao, S.-H., Mei, J., Xu, J., Fan, D.-P., Zhang, R.-G., & Cheng, M.-M. (2021). Jcs: An explainable covid-19 diagnosis system by joint classification and segmentation. *IEEE Transactions on Image Processing*, *30*, 3113–3126.
- Xu, C., Li, Q., Zhou, M., Zhou, Q., Zhou, Y., & Ma, Y. (2022). Rgb-t salient object detection via cnn feature and result saliency map fusion. *Applied Intelligence*, *52*(10), 11343–11362.
- Yuan, Y., Liu, W., Gao, P., Dai, Q., & Qin, J. (2024). Unified unsupervised salient object detection via knowledge transfer. arXiv:2404.14759.
- Zhang, J., Fan, D.-P., Dai, Y., Anwar, S., Saleh, F. S., Zhang, T., & Barnes, N. (2020a). Uc-net: Uncertainty inspired rgb-d saliency detection via conditional variational autoencoders. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 8582–8591).

- Zhang, P., Xu, M., Zhang, Z., Gao, P., & Zhang, J. (2023). Feature aggregation with transformer for RGB-t salient object detection. *Neurocomputing*, 546, 126329.
- Zhang, Q., Huang, N., Yao, L., Zhang, D., Shan, C., & Han, J. (2019). Rgb-t salient object detection via fusing multi-level cnn features. *IEEE Transactions on Image Processing*, 29, 3321–3335.
- Zhang, Q., Xiao, T., Huang, N., Zhang, D., & Han, J. (2020b). Revisiting feature fusion for RGB-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 31(5), 1804–1818.
- Zhang, Y., Shen, Z., & Jiao, R. (2024). Segment anything model for medical image segmentation: Current applications and future directions. *Computers in Biology and Medicine*, 171, 108238.
- Zhao, J.-X., Liu, J.-J., Fan, D.-P., Cao, Y., Yang, J., & Cheng, M.-M. (2019). Egnnet: Edge guidance network for salient object detection. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 8779–8788).
- Zhou, H.-Y., Liu, A.-A., Nie, W.-Z., & Nie, J. (2019). Multi-view saliency guided deep neural network for 3-d object retrieval and classification. *IEEE Transactions on Multimedia*, 22(6), 1496–1506.
- Zhou, W., Guo, Q., Lei, J., Yu, L., & Hwang, J.-N. (2021a). Ecffnet: Effective and consistent feature fusion network for rgb-t salient object detection. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(3), 1224–1235.
- Zhou, W., Sun, F., Jiang, Q., Cong, R., & Hwang, J.-N. (2023a). Wavenet: Wavelet network with knowledge distillation for RGB-t salient object detection. *IEEE Transactions on Image Processing*, 32, 3027–3039.
- Zhou, W., Zhu, Y., Lei, J., Wan, J., & Yu, L. (2021b). Apnet: Adversarial learning assistance and perceived importance fusion network for all-day rgb-t salient object detection. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 6(4), 957–968.
- Zhou, W., Zhu, Y., Lei, J., Yang, R., & Yu, L. (2023b). Lsnet: Lightweight spatial boosting network for detecting salient objects in rgb-thermal images. *IEEE Transactions on Image Processing*, 32, 1329–1340.